Intent-Driven Spatial Search: A Hybrid NL-to-SQL Architecture for Conversational Geographic Queries

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***Abstract*—Mainstream digital mapping engines exhibit catastrophic spatial prioritization failures when confronted with descriptive, intent-driven queries such as *"quiet places near me."* Their reliance on deterministic keyword filtering against business-name indexes leads to results separated from the user by hundreds of kilometers, breaking the implicit spatial constraint of locality. This work proposes an intent-driven spatial search architecture that translates conversational queries into executable PostGIS SQL via a specialized Text-to-SQL model. The system couples a denormalized OpenStreetMap-derived geographic database for the Pilani region with a hand-authored Semantic Intent Ontology that maps colloquial vocabulary ("quiet," "late-night," "study spot") to formal geographic tags. A custom Hybrid Evaluation Framework, fusing Abstract Syntax Tree (AST) structural matching with execution-based F1 semantic scoring, is introduced to overcome the false positives and false negatives inherent in academic baselines. On a 50-query Golden Benchmark Suite stratified by difficulty, the cloud-served Defog SQLCoder-8B reference configuration achieves an 84.0% pass rate (avg. structural 0.827, avg. semantic F1 0.784). To achieve full offline operation, the model was subsequently re-deployed via llama.cpp with 4-bit GGUF quantization and a 2B-parameter Kimi grammar parser; this configuration regressed in accuracy due to quantization-induced precision loss in schema linking and the limited grammatical capacity of the smaller parser, motivating the future-work agenda. The work establishes that intent-based NL-to-SQL routing is a viable replacement for flat keyword matching in localized spatial search.**

***Index Terms*—Conversational AI, geographic information systems, intent recognition, llama.cpp, natural language interfaces, OpenStreetMap, PostGIS, quantization, semantic ontology, spatial databases, Text-to-SQL.**

I. INTRODUCTION

A. Background

Digital mapping and navigational systems have become integral to everyday life, yet their underlying search architectures remain fundamentally constrained when processing human-centric, conversational queries. Most mainstream mapping engines rely heavily on deterministic keyword filtering. While these systems excel at locating standardized addresses or exact proper nouns, they suffer from catastrophic spatial prioritization failures when users input descriptive, intent-driven queries.

A canonical example is the *Local Intent Failure*. When a user located in Pilani searches for *"quiet places near me,"* current systems fail to comprehend *"quiet"* as a desired spatial attribute. Instead of parsing the semantic intent, the engine defaults to a rigid, exact-keyword match against its database of registered business names, surfacing results such as a venue named "Quiet Place" in HSR Layout, Bengaluru, more than 1,000 km from the user. The explicit *"near me"* constraint is silently abandoned in favor of textual relevance. Such failures actively erode user trust and demonstrate that the inability to process conversational spatial queries is not a superficial UI issue but a deep architectural problem.

In the Indian context this limitation is especially pronounced. Spatial references are rarely confined to perfectly structured street addresses; they are highly descriptive, informal, and contextual. Users frequently search for places based on utility, vibe, or immediate physical needs—"late-night study cafes," "safe parks for kids," "uncrowded restaurants." When mapping engines treat these nuanced multi-dimensional requests as flat text strings to be matched against formal business registrations, the digital map becomes effectively useless for intuitive local exploration.

Addressing this gap requires a paradigm shift from rigid keyword pipelines to advanced, intent-driven spatial modeling. The system must distinguish between a user looking for a place named "Quiet" and a user seeking the attribute of "quietness" within a tightly bounded local radius. This project pursues that fundamental architectural transformation, mapping human-centric vocabulary to underlying geographic realities.

B. Motivation

Despite the ubiquity of digital maps, their effectiveness remains deeply uneven for natural, human-centric spatial queries. There is a significant performance gap between structured address-based searches and descriptive, intent-driven requests. This disparity is driven by current architectures being optimized for exact noun matching rather than understanding semantic characteristics or environmental attributes of a location.

Existing engines struggle particularly in regions like India, where spatial references are unstructured, heavily reliant on informal landmarks, and expressed through colloquial descriptions rather than grid-based coordinates. Because current engines lack intent modeling, they exhibit poor performance when users search for places based on utility or vibe; relevant local spaces are ignored simply because their formal business registrations do not contain the user's descriptive keywords.

Solving this problem does not require expanding existing keyword databases—a brute-force approach that fails to address the root semantic disconnect. It motivates the exploration of intelligent, intent-driven architectural alternatives. By dynamically mapping conversational queries to spatial attributes, accurate local search can be achieved without rigid exact-match filtering.

C. Objective

The primary objective is to architect an intelligent, intent-driven spatial search engine that bridges the structural gap between natural human language and formal geographic databases. Rather than processing inputs as flat text strings, the architecture deconstructs multi-dimensional requests, mapping human-centric vocabulary to underlying environmental attributes and implicit spatial constraints. The work is organized around four technical pillars:

- Geospatial database and ontology construction: a denormalized PostgreSQL/PostGIS framework integrated with a Semantic Intent Ontology that maps colloquial expressions to structured geographic attributes.
- Dynamic intent capturing and NL-to-SQL translation: optimization of specialized Text-to-SQL models (Defog SQLCoder) to extract explicit and implicit intents and translate them into executable, strictly bounded spatial commands.
- Rigorous hybrid evaluation: a custom framework that fuses AST structural parsing with live database execution to validate accuracy against real-world spatial entropy.
- Conversational interaction layer: a multi-modal interface that manages spatial context across follow-up turns and synthesizes raw database outputs into localized navigational dialogue.

II. RELATED WORK AND COMPARISONS

A. Traditional Geocoding and Commercial Mapping Engines

The most common approach to spatial search relies on hierarchical indexing and information-retrieval (IR) techniques. Nominatim, the primary search engine for OpenStreetMap (OSM), uses a structured spatial index over country, state, city, street, and house number. While precise for exact addresses and administrative boundaries, its architecture is strictly deterministic and semantically blind: it fails completely on descriptive queries such as “quiet places” or “late-night cafes.”

Industry-standard platforms such as Google Maps and MapmyIndia/Mappls combine massive proprietary Point-of-Interest (POI) databases with text-ranking algorithms based on TF-IDF or Elasticsearch. They excel at locating registered businesses and incorporating reviews, but their architecture is flawed for natural language. They suffer from an Exact-Match Bias, weighting exact word matches in business titles over geographic proximity, so a search for “quiet places near me” in Pilani may return “Quiet Place” in Bengaluru. Queries are treated as flat text strings rather than as multi-dimensional intents combining utility, environment, and spatial bounds.

B. Semantic POI Recommendation Systems

Recent academic work uses semantic embeddings and graph neural networks to model relationships between user preferences, prior behavior, and venue tags. These systems capture category synonyms and the “vibe” of a location that traditional geocoders miss. However, they exhibit Weak Spatial Enforcement: they rank results by semantic text similarity first and treat spatial proximity as a secondary signal, struggling to enforce hard geographic boundaries. They are also notoriously difficult to integrate directly with raw relational spatial databases such as PostGIS for low-latency querying.

C. Research Gaps and Proposed Advantages

A review of current spatial search architectures reveals three major research gaps:

- Exact-Match Bias (Name vs. Attribute): systems prioritize exact matches in formal business titles, leading to catastrophic routing failures when users search for environmental attributes.
- Deterministic Filtering: an inability to dynamically map colloquial human-centric needs (e.g., “uncrowded,” “safe,” “study spot”) to formal geographic tags.
- Decoupled Architecture: a fundamental disconnect between natural language processing and the underlying relational spatial database, in which spatial constraints are abandoned in favor of semantic text matching.

This project addresses these gaps by shifting the architecture from rigid keyword filtering to intent-based spatial routing. Conversational queries are translated into structured multi-dimensional filters; descriptive language is deconstructed and mapped against environmental and utility attributes via a semantic ontology. Because natural language is mapped directly into executable PostGIS SQL, semantic intent is natively understood by the database, enabling complex descriptive constraints and strict geographic boundaries to be evaluated simultaneously at the data layer.

III. PROPOSED TECHNIQUE AND ALGORITHM

A. Architectural Evolution and Model Selection

The initial conceptualization of the NL-to-SQL pipeline used large API-dependent general-purpose models, beginning with Llama-3-70B-Versatile. While capable, this approach was unviable for a scalable mapping engine due to prohibitive API latency, token cost, and dependency on proprietary keys. The architecture was pivoted toward Defog SQLCoder, an open-source state-of-the-art model fine-tuned for Text-to-SQL. Local deployment is critical for spatial systems: it ensures data privacy, eliminates network-induced latency during query generation, and allows deep integration with local geographic databases.

B. Evaluation Strategy Formulation

Evaluating NL-to-SQL systems is notoriously difficult, particularly in the geographic domain. A query can be syntactically correct but logically flawed, or syntactically different from a gold standard but logically perfect. In highly denormalized spatial databases—where a single location may carry overlapping tags as a cafe, library, and university—traditional benchmarking falls short. Two leading academic frameworks were analyzed.

1) Component-Based Structural Matching (Spider):

The Spider methodology [1] avoids fragile string matching by parsing predicted and gold queries into structural components (SELECT, WHERE, GROUP BY, ORDER BY) and applying "Exact Set Match" over their columns and operators. While structured, it penalizes models for writing valid queries that achieve the correct result via different logic, producing high false-negative rates on spatial schemas where multiple filtering paths are equally valid.

2) Execution-Based Semantic Matching (Test Suite):

The Test Suite approach [2] runs the predicted query against distilled databases and accepts it if its output rows match the gold output. In highly denormalized geographic databases, however, a poorly constructed query may accidentally return the correct result due to data sparsity—for example, a vague query for "all large buildings" coincidentally matching the single hospital row returned by the gold query. This produces high false-positive rates.

C. The Proposed Hybrid Evaluation Framework

To eliminate the respective false positives and false negatives of the academic baselines, this work proposes a Hybrid Evaluation Framework that fuses AST-based structural testing with execution-based semantic testing.

1) Golden Dataset Creation:

A custom benchmark of 50 spatial queries was hand-authored to reflect Indian-context user intents ("Find late-night cafes near me," "Quiet study spots," "Places that accept UPI"). Each entry contains the natural-language question, a difficulty class (Easy, Medium, Hard, Extra), a target intent cluster, and a manually verified Gold SQL query authored to PostGIS standards.

2) AST-Based Structural Testing:

The framework uses sqlglot to parse both the predicted and gold queries into Abstract Syntax Trees, mathematically normalizing table aliases and extracting SELECT columns, FROM tables, WHERE columns, WHERE operators (ILIKE, =, IS NULL), ORDER BY clauses, and LIMIT constraints. A structural score is calculated from the intersection of these sets, ensuring the model is logically querying correct spatial tags rather than hallucinating columns.

3) Execution-Based Semantic Testing:

The framework connects to the local PostgreSQL/PostGIS database via SQLAlchemy, dynamically strips arbitrary LIMIT constraints from both queries to prevent alphabetical truncation from causing false mismatches, executes both, and extracts the resulting geographic entities into normalized sets. An F1-score is computed over predicted vs. gold entity sets. Given the entropy of real-world spatial data, $F1 \geq 0.50$ is taken as the threshold for a successful semantic match.

4) Final Hybrid Verification:

The final algorithm merges structural and semantic scores into a definitive classification: PASS (structural ≥ 0.50 and semantic ≥ 0.50); STRUCT_ONLY (correct schema but incorrect data—catches false positives of pure execution scoring); SEM_ONLY (correct results via unconventional SQL—catches false negatives of Spider matching); or FAIL (both fail, indicating complete translation breakdown).

D. Conversational Interaction Layer

Exposing raw database execution to end users is impractical, so the architecture incorporates a Conversational Interaction Layer that operates over three architectural functions. First, a dynamic input parser maintains a spatial state across turns: when the user follows up with "what about ones open late?" the layer retains the original constraints (quiet, study) and appends the new utility constraint before re-routing the prompt to the model. Second, a data-to-dialogue translator synthesizes raw PostGIS rows into fluid responses such as "I found three quiet study spaces within 2 km; the closest is the Central Library, but The Reading Room is open until midnight." Third, multi-modal spatial synchronization couples the dialogue with the visual map: when the system mentions a venue it simultaneously plots its coordinates, adjusts the bounding box, and highlights the route, providing tight visual feedback for conversational intent.

IV. DATASET

A. The Spatial Database (Geographic Ground Truth)

Underlying data was sourced from OpenStreetMap and ingested into a locally hosted PostgreSQL database with the PostGIS extension, focused on the Pilani region. Rather than clean normalized relational tables, the database is intentionally a denormalized flat table (pilani_places) that mirrors the chaotic, tag-heavy nature of real-world GIS systems and forces the engine to infer constraints from inconsistent or asymmetrical data shapes. The schema groups data into core identifiers (OSM IDs, names, multilingual tags), geospatial data (PostGIS geometry columns for distance and bounding-box calculations), primary place categories (amenity, building, shop, leisure, sport), and granular utility/attribute tags (opening_hours, indoor, lit, surface, payment:upi).

B. Semantic Intent Ontology

Formal geographic databases do not store subjective experiences or utility-based colloquialisms, so a structured mapping dictionary was authored as the connective tissue between user vocabulary and the database schema. Examples include: a "Late Night" intent maps to opening_hours ILIKE '%24/7%' OR '%22:%' OR '%23:%' OR '%00:%'; a "Study" intent maps to amenity ILIKE '%library%' OR building ILIKE '%university%'; a "Photogenic / Vibe" intent maps to a combination of tourism, historic, artwork_type, and man_made columns; a "Digital Payment" intent maps to the boolean flag payment:upi = 'yes'. The ontology was injected into the system prompt, giving the translation model the domain knowledge to route subjective queries to objective data points.

C. The Golden Benchmark Suite

A custom Golden Benchmark Suite of 50 hand-crafted queries was authored to test intent-based spatial routing. Queries are stratified across four difficulty levels (Easy, Medium, Hard, Extra Hard); each is paired with a manually verified, structurally clean Gold SQL query. This dataset serves as the empirical ground truth for the Hybrid Evaluation Framework, enabling precise measurement of the system's ability to translate messy colloquial requests into strict executable geographic rules.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

Two configurations were evaluated. Configuration A is the cloud-served reference: Defog SQLCoder-8B accessed through hosted inference, processing the 50-query Golden Benchmark Suite. Configuration B is the fully offline deployment: the same SQLCoder weights re-deployed locally via llama.cpp with 4-bit GGUF quantization, paired with a 2B-parameter Kimi grammar parser used as the conversational front end. Both configurations were graded by the Hybrid Evaluation Framework (v4.1), which independently computes an AST-based structural score (via sqlglot) and an execution-based semantic F1 against the local PostGIS database. A query passes only if structural ≥ 0.50 and semantic F1 ≥ 0.50. To ensure rigorous semantic validation the framework dynamically strips arbitrary LIMIT constraints, comparing whole result sets rather than truncated samples.

B. Evaluation and Results

Configuration A (cloud SQLCoder-8B) demonstrated robust translation, mapping conversational vocabulary to structured constraints in the vast majority of test cases. Of the 50 queries: 42 PASS (84.0%), 1 FAIL (2.0%), 7 STRUCT_ONLY (14.0%), 0 SEM_ONLY, 0 NO_PRED. The average structural score was 0.827 and the average semantic F1 was 0.784. The high structural score indicates the model excels at schema linking—consistently identifying correct tables, columns, and operators without hallucinating invalid schema. The 7 STRUCT_ONLY cases produced syntactically valid SQL that matched the database structure but failed semantic execution; these were partial matches rather than complete failures, falling below the 0.50 F1 threshold. This validates the necessity of the hybrid framework: pure structural scoring would have marked these false positives, while a top-5 sampled execution comparison would have marked them as passes.

TABLE I

CONFIGURATION A PERFORMANCE BY QUERY COMPLEXITY

Difficulty	Total	Pass	Pass %	Struct	Sem F1
Easy	19	17	89.5	0.851	0.842
Medium	16	14	87.5	0.844	0.825
Hard	12	9	75.0	0.750	0.708
Extra	3	2	66.7	0.889	0.500
Overall	50	42	84.0	0.827	0.784

Performance degrades gracefully across complexity tiers. The system excels at direct attribute lookups (mapping "parks" to leisure ILIKE '%park%'), with Easy and Medium pass rates near 90% and F1 above 0.80. On Hard queries (75% pass) the model struggles with asymmetrical schemas, such as filtering for boolean flags lacking direct column equivalents and constructing strict negations (e.g., buildings that are not places of worship), reflected in the structural-score drop to 0.750. On Extra Hard queries the structural score rebounds to 0.889 but F1 lands exactly on the 0.500 threshold: the model writes well-formed SQL but fails to capture the entirety of the required semantic subset when parsing highly informal multi-part keyword strings.

C. Offline Deployment Update (Configuration B)

To eliminate the remaining cloud dependency and achieve full offline operation on a campus laptop, the SQLCoder weights were re-deployed via llama.cpp with 4-bit GGUF quantization, and a 2B-parameter Kimi grammar parser was placed in front of the model as the conversational front end. The motivation was threefold: removing API latency, eliminating per-token cost, and ensuring data privacy in the absence of campus Wi-Fi. The new pipeline runs entirely on CPU/GPU local resources with substantially reduced VRAM consumption due to INT4 weight quantization.

Two consequent regressions were observed against Configuration A. First, INT4 quantization introduced precision loss in the model's schema-linking layer: the quantized network occasionally produced near-correct but subtly wrong column names or malformed JOIN clauses on Hard and Extra Hard queries—classes already sensitive to logical-grouping errors. Second, the 2B-parameter Kimi parser exhibited weaker grammatical decomposition than the original conversational front end, producing under-specified prompts that propagated into the Text-to-SQL stage and amplified the partial-match problem already documented in Configuration A. The net effect was an aggregate accuracy reduction concentrated in the Hard and Extra tiers; Easy and Medium tiers remained largely robust because their structural simplicity tolerates moderate precision loss. These observations are consistent with the general literature on aggressive quantization of code- and SQL-specialized LLMs, and they motivate the future-work agenda described in Section VII.

D. Error Analysis

Analysis of the failure and STRUCT_ONLY cases reveals two architectural limitations that warrant optimization, both amplified under Configuration B:

- Over-Aggressive Logic Grouping (the Partial-Match Problem). The majority of missed semantic targets occur because the model applies AND logic where implicit OR logic is contextually required. When a user asks for "volleyball or table-tennis courts," the model occasionally searches for venues possessing both tags simultaneously. While this returns valid partial matches, comparing whole result sets exposes that it artificially narrows the geographic bounding box, dragging F1 below 0.50.

- String-Matching Fragility. On Extra-difficulty queries the model occasionally relies on overly rigid string matching. If a user asks for “Meera Bhawan girls hostel,” the database may only list it as “Meera Bhawan.” The semantic ontology resolves generic categories such as “quiet” or “study,” but entity resolution for unregistered colloquial nicknames of specific buildings remains incomplete, reducing semantic recall.

Despite these edge cases, the experiment establishes that replacing flat text indexing with intent-based NL-to-SQL routing yields highly accurate, strictly bounded, semantically aware spatial results in 84.0% of Configuration A test cases.

VI. CONCLUSION AND FUTURE WORK

This work demonstrates that the catastrophic spatial prioritization failures of mainstream mapping engines on conversational queries are not a UI defect but a deep architectural one, and that an intent-driven NL-to-SQL routing pipeline coupled with a semantic ontology and a hybrid AST-plus-execution evaluation framework can replace flat keyword indexing for localized search. The cloud-served reference configuration achieves 84.0% pass rate on a stratified 50-query benchmark; the fully offline configuration based on llama.cpp 4-bit quantization and a 2B Kimi parser is functional but regresses in accuracy due to quantization-induced precision loss and weaker grammatical front-end parsing.

Future work focuses on closing this regression gap: (i) selectively de-quantizing the schema-linking layers of the offline SQLCoder while keeping general layers in INT4 to recover Hard-tier accuracy; (ii) replacing the 2B Kimi parser with a 7B-class on-device front end to reduce upstream prompt under-specification; (iii) refining the prompt to enforce OR-grouping in cases of disjunctive utility intents; (iv) expanding the Semantic Intent Ontology with localized building aliases and informal hostel nicknames to improve entity resolution on the Extra-difficulty tier; and (v) optimizing end-to-end latency between user input and PostGIS response so the offline system is competitive with the cloud reference for real-time campus use.

VII. WORK UPDATE

A. Work Completed

The foundational architecture has been designed, implemented, and rigorously evaluated. The project has moved beyond the theoretical phase into a functional end-to-end routing prototype. Specific milestones include: a robust geographic ground-truth database for the Pilani region deployed on PostgreSQL/PostGIS, including manual ingestion of OSM data and creation of the Semantic Intent Ontology that maps colloquial descriptors to formal database tags; a stratified 50-query Golden Benchmark Suite covering Food, Academics, Residential, and Logistics intent clusters; identification and integration of Defog SQLCoder-8B as the optimal translation engine; development and execution of the Hybrid Evaluation Framework, achieving 84.0% pass rate against a 0.50 F1 threshold; and deployment of a functional prototype in which a generic LLaMA model serves as the conversational front end, demonstrating end-to-end routing of natural language through the specialized SQL engine. As an architectural upgrade, the SQLCoder model has additionally been re-deployed offline via llama.cpp with 4-bit GGUF quantization paired with the Kimi 2B grammar parser, eliminating cloud dependency at a measured accuracy cost discussed in Section V-C.

B. Work Remaining

The transition from prototype to specialized production system requires several developments. The current LLaMA-based front end will be replaced with a custom Conversational Interaction Layer that maintains spatial state across follow-up turns; an advanced data-to-dialogue synthesis module will explain why specific results were chosen rather than emitting raw rows; prompt refinement will address the 16% partial-match rate, particularly the Over-Aggressive Logic Grouping pattern; the ontology will be expanded for entity resolution of localized nicknames (“Meera Bhawan girls hostel”); and final latency optimization will minimize the time between user input and PostGIS response. In parallel, the offline configuration will undergo the targeted-precision and parser-upgrade interventions outlined in Section VI to recover accuracy lost to aggressive quantization.

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